

# Federico Mazzone

Cork, Ireland · mazzoni.federico1@gmail.com · [LinkedIn Profile](#) · [GitHub Portfolio](#) · [Scholar](#)

**AI Engineer** with experience building and deploying production LLM, RAG, and automation systems on AWS. Delivered customer-facing and internal AI products that increased engagement and reduced workflows from days to hours. Background in NLP and machine learning, with 8 peer-reviewed AI papers. EU citizen based in Ireland.

## EXPERIENCE

### AI Engineer

*Marriott*

August 2026 – Present

*Cork, Ireland*

- Recently joined the AI Engineering team, contributing to the development and deployment of AI-driven solutions.

### AI Engineer

*Domotz*

May 2024 – July 2026

*Dublin, Ireland*

- **Chatbot:** Engineered a FastAPI RAG-based chatbot (using Qdrant, LangChain, PostgreSQL) deployed on AWS, boosting customer engagement by 65% and serving 100+ daily users. Built multiple knowledge bases and UIs.
- **Dynamic MCP:** Developed an MCP server acting as a semantic proxy across multiple MCP servers, dynamically providing only the tools and skills relevant to each user's task via hybrid semantic search.
- **Conversation Intelligence:** Engineered an in-house Gong alternative for Sales & CS teams, featuring zero-shot LLM classification, summarization, and RAG-based chat over thousands of emails, tickets, and call transcripts.
- **Automation Systems:** Built automation systems that reduced sales, finance, and marketing workflows from days to hours, including a Slack sales bot and invoice processors handling hundreds of customers.
- **Anomaly Detection:** Combined traditional ML and LLM reasoning to detect network anomalies in time series.

### PhD Research Fellow in AI

*University of Pisa / National Research Council of Italy*

Nov. 2022 – May 2024

*Pisa, Italy*

- **Human-in-the-Loop:** Built a hybrid AI-human feedback loop system, boosting label accuracy up to 50%.
- **NLP:** Developed a library to quantify trust in social networks using embedding similarity and information density.
- **Algorithmic Fairness:** Designed algorithms to improve fairness in social networks and tabular ML models.
- Published 8 peer-reviewed articles, managed multiple concurrent projects from concept to validation, mentored university students, presented research at conferences, and organised the *Law, Society, and AI* workshops.
- Conducted 1.5 years of PhD research, before transitioning to industry for practical applications of AI.

### Freelance Data Analyst

*Self-Employed*

2022

*Pisa, Italy*

- **Recommendation System:** Developed content-based and collaborative filtering recommendation systems.
- **Data Collection:** Created datasets by scraping Wikipedia and independent sites with Selenium/Beautiful Soup.
- **Bioinformatics:** Explored methods in *R* to identify potential biomarkers in Rett syndrome data.

## PERSONAL PROJECTS

**GAVEL:** Ethical evaluation of AI model behaviour and decision-making using Game Theory (WIP).

**iRAG:** Temporal-aware RAG that learns from human decisions to enable trusted autonomy. Demo for CRMs [here](#).

**StockAgent:** Multi-agent AutoGen system for stock market analysis with a custom MCP yfinance server.

**HINTT:** Real-time OCR+translation tool sending screenshots to OpenAI/Ollama. Employed tkinter and mss.

**Simple Palette Maker:** Streamlit app to colorize black-and-white GameBoy palettes. 1k+ users.

## SKILLS

**AI/ML Stack:** LLMs (LangChain, LangGraph, AutoGen, Bedrock), RAG, HuggingFace, SK-Learn, PyTorch, RiverML

**Backend & Deployment:** FastAPI, Streamlit, Docker, AWS (EC2, S3, CloudWatch, Terraform), PostgreSQL, Qdrant

**Programming:** Python, JavaScript, R

**Platforms:** Git, Jira, BitBucket, Snowflake, Zendesk, Salesforce, n8n, OpenAI, Claude, Amazon Bedrock, Azure

**Languages:** Italian (Native), English (C1), German (A2)

## EDUCATION

**University of Southern Denmark**

*Summer School on Deep Learning*

2023

*Odense, Denmark*

**University of Pisa**

2020 – 2022

*Master's Degree in Digital Humanities (Data Science and NLP) — 110/110 with honours (GPA: 4/4) Pisa, Italy*

- Relevant Courses: Data Mining, Social Network Analysis, Text Analytics, Distributed Data Analysis and Mining, Semantic Web, Laboratory of Data Science, Applied Linguistics, Computational Linguistics.

**University of Pisa**

2014 – 2017

*Bachelor's Degree in Philosophy — 110/110 with honours (GPA: 4/4) Pisa, Italy*